

CAP BEAR, France

WMO: 077490

Lat: 42.5162N

Lon: 3.1333E

Elev: 285

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.1	38.0	18.9	14.4	40.9	22.6	17.1	41.9	69.2	45.8	64.0	46.8	33.8	320	0.750

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	84.8	69.1	82.7	68.8	80.8	68.4	74.3	78.8	72.9	77.7	71.7	76.9	14.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	123.4	76.8	71.4	117.2	75.8	69.9	111.3	75.0	37.9	78.8	36.7	77.7	35.6	76.7	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	60.0	53.9	48.3	DB	32.4	89.4	3.2	2.7	30.1	91.3	28.2	92.9	26.4	94.4	24.1	96.3
(5)				WB	28.6	76.4	3.1	1.9	26.3	77.7	24.5	78.8	22.7	79.8	20.5	81.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	48.9	49.3	53.3	57.3	63.2	70.6	74.8	74.7	68.9	63.4	55.6	51.1	(6)
		DBStd	10.37	5.16	5.42	4.78	4.53	4.66	4.75	3.97	3.70	4.11	4.71	5.34	5.12	
		HDD50	238	81	67	22	3	0	0	0	0	1	16	48		
		HDD65	2438	498	438	364	235	89	7	0	0	84	282	430		
		CDD50	4255	48	49	124	221	409	617	770	767	566	416	185	83	
		CDD65	979	0	0	0	4	32	174	305	302	127	34	1	0	
		CDH74	4219	0	0	0	3	63	635	1664	1554	264	36	0	0	
		CDH80	600	0	0	0	0	5	87	259	229	18	2	0	0	
(14)	Wind	WSAvg	19.9	22.3	23.0	20.7	20.5	19.5	17.9	18.6	17.5	18.2	18.7	22.0	20.6	(14)
(15)	Precipitation	PrecAvg	25.0	2.4	1.8	2.0	2.2	2.1	1.2	0.6	1.1	2.3	3.7	3.1	2.4	(15)
(16)		PrecMax	37.7	7.8	5.2	6.6	8.4	5.0	5.6	1.7	2.8	6.8	10.5	9.4	8.6	(16)
(17)		PrecMin	17.8	0.0	0.0	0.0	0.2	0.3	0.1	0.1	0.7	0.2	0.5	0.1		(17)
(18)		PrecStd	6.0	2.1	1.4	1.8	1.9	1.5	1.2	0.4	0.7	1.5	2.6	2.8	2.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	63.6	63.8	68.9	72.4	79.7	85.6	87.4	89.0	82.7	78.3	69.2	64.9		(19)
(20)		0.4% MCWB	53.3	51.9	54.7	58.3	64.8	68.4	70.0	71.0	67.2	64.1	59.9	55.0		(20)
(21)		2% DB	60.5	60.2	64.7	68.5	75.2	82.3	84.7	85.0	79.1	73.9	66.3	61.7		(21)
(22)		2% MCWB	52.0	51.5	53.8	57.2	62.9	68.0	69.2	69.8	66.8	64.2	59.0	53.8		(22)
(23)		5% DB	58.2	58.1	61.8	66.1	71.9	79.8	82.8	82.3	76.9	71.5	64.4	59.7		(23)
(24)		5% MCWB	51.0	50.4	52.5	56.4	61.1	67.1	68.8	69.6	66.2	63.6	58.5	53.1		(24)
(25)		10% DB	56.2	56.2	59.5	63.9	69.7	77.4	81.0	80.3	75.0	69.3	62.8	57.9		(25)
(26)		10% MCWB	49.7	49.7	51.9	55.4	60.3	66.3	68.5	68.9	65.5	62.7	57.3	52.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	55.6	54.9	57.5	61.3	66.8	73.7	75.8	76.4	72.2	68.6	63.4	57.6		(27)
(28)		0.4% MCDB	60.7	58.8	63.9	67.7	74.6	77.6	80.6	80.5	76.0	72.2	65.2	61.0		(28)
(29)		2% WB	53.7	53.5	55.7	59.4	64.9	71.3	73.9	74.7	70.5	67.0	61.7	55.8		(29)
(30)		2% MCDB	58.0	57.4	60.6	65.3	71.3	76.6	78.8	79.2	74.6	70.3	64.5	59.8		(30)
(31)		5% WB	52.2	52.1	54.6	58.1	63.5	69.8	72.7	73.4	69.2	65.8	59.8	54.4		(31)
(32)		5% MCDB	56.7	56.3	58.9	63.5	69.3	76.3	78.1	77.9	73.6	68.7	63.3	58.6		(32)
(33)		10% WB	50.7	50.9	53.5	56.8	62.2	68.4	71.3	72.2	67.9	64.6	58.2	52.8		(33)
(34)		10% MCDB	55.3	55.3	57.4	61.8	67.4	75.2	77.4	76.9	72.8	67.9	61.7	57.1		(34)
(35)	Mean Daily Temperature Range	MDBR	8.1	8.6	8.8	9.2	9.3	9.8	10.0	9.8	9.3	8.3	8.1	8.0		(35)
(36)		5% DB MCDBR	10.0	11.0	11.5	11.5	11.8	12.0	11.6	11.2	10.9	10.2	9.3	9.6		(36)
(37)		5% WB MCWBR	7.1	7.4	7.2	6.9	6.8	6.4	7.0	7.4	7.4	6.9	6.7	6.7		(37)
(38)		5% WB MCDBR	9.1	9.5	9.5	10.1	10.1	10.0	10.1	9.7	9.4	8.2	7.7	8.3		(38)
(39)		5% WB MCWBR	7.0	7.3	6.8	6.9	6.7	6.5	7.1	7.6	7.3	6.2	6.5	6.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.329	0.371	0.399	0.411	0.419	0.420	0.416	0.396	0.376	0.339	0.313		(40)
(41)		taud	2.521	2.482	2.382	2.313	2.330	2.342	2.352	2.365	2.404	2.458	2.505	2.527		(41)
(42)		Ebn at Noon	264	278	276	276	274	272	270	267	265	256	251	253		(42)
(43)		Edh at Noon	23	27	34	39	40	39	38	37	33	28	23	21		(43)
(44)	All-Sky Solar Radiation	RadAvg	580	873	1256	1588	1859	2121	2116	1858	1437	975	649	516		(44)
(45)		RadStd	61	86	102	128	126	102	101	93	81	83	66	49		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83

Nomenclature: See separate page